CAPSTONE PROJECT PITCH

Next-Gen Music Recommender

on real Yandex listening data (Yambda)

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HSE Master's · NNDL · Module 2

Real-world operational data (+2 bonus)

github.com/ak1232320/nndl_capstone_2026

01 / PROBLEM STATEMENT

Predict the next track for a streaming user

What we are solving

Given a 10 M-track catalogue, predict the next track a user will actually want to hear — the central decision a streaming product makes every few minutes.

Why now

- ≈ 70 % of streaming users struggle to discover new music — bad recommendations are the dominant churn driver.
- The streaming market is saturated; subscriber growth has plateaued, so retention is the #1 commercial metric.
- A few % lift in next-track ranking separates catching up from leading.

Yambda · 4.79 B real listening events from Yandex Music

Source

- Yambda by Yandex on Hugging Face — released May 2025 (arXiv:2505.22238, RecSys '25).
- 1 M users · 9.39 M tracks · 4.79 B events **(we train on the 50M slice)**.

Why it is strong for this task

- Three signal layers: explicit (likes/dislikes), implicit (plays/skips/replays), and 128-d audio embeddings (82 % of tracks).
- An `is_organic` flag on every event separates user-driven from recommender-driven plays — trains without the recommender-bias loop.
- Real production distributions: heavy-tail catalogue, real cold-start, real session structure.

Real-world operational data → qualifies for the +2 bonus, target = 10 / 10.

03 / MODELING APPROACH

Hybrid: SASRec sequence model + audio-content tower

A · SASRec — sequence

- Causal-attention Transformer over the user's listening history (structurally a small GPT).
- Learns transition patterns and predicts the next track from the sequence so far.

B · Content tower

- Scores tracks by audio-taste similarity on Yandex 128-d embeddings — complementary to the sequence model.

Fusion & why this approach

- Outputs fused with a **validation-tuned weight (β)** — late, not joint, fusion (joint overfit).
- Pure CF loses sequence; pure content ignores behaviour — the hybrid handles both.

NDCG@10 — ranking quality at the top of the list

Primary metric

NDCG@10 under the dataset's Global Temporal Split, ranking over the full 629 k-item catalogue (the same protocol as the Yandex paper).

Results (our harness)

- Best baselines: ItemKNN 0.071 · **SASRec 0.0735** (98 % of the paper, beats all baselines).
- **Hybrid: +6–10 % over SASRec** — robust, positive on every split and run.

Expected business impact

- +6–10 % better next-track ranking → lever on listening time and churn.
- Delivered on free data + one free GPU (near-zero infrastructure cost).

05 / SUMMARY & CALL TO ACTION

What we ship

Real Yandex listening data (Yambda-50M) → hybrid neural recommender: SASRec + audio-content tower, fused with a validation-tuned weight → predict the next track. **SASRec beats every baseline and reproduces the paper to 98 %; late fusion adds a robust +6–10 % NDCG@10** — positive on every split and run.

What it took / how to reproduce

- Yambda dataset on Hugging Face — public, free.
- One free Kaggle T4 (16 GB) GPU — no paid infrastructure.
- Reproducible: **github.com/ak1232320/nndl_capstone_2026** — one notebook **RUN_ALL.ipynb** (Run All).